

Lab 10: Programs using CUDA memory types and synchronization

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```
6CSEC2@debian:~/Documents/230905252/Lab10$ nvcc -o exe/q0 q0_matMult_tiled.cu
6CSEC2@debian:~/Documents/230905252/Lab10$ nvcc -o exe/q1 q1_matMult.cu
6CSEC2@debian:~/Documents/230905252/Lab10$ nvcc -o exe/q2 q2_conv1D_constant.cu
6CSEC2@debian:~/Documents/230905252/Lab10$ nvcc -o exe/q3 q3_conv1D_tiled.cu
6CSEC2@debian:~/Documents/230905252/Lab10$ nvcc -o exe/q4 q4_shoppingMall.cu
```

```
6CSEC2@debian:~/Documents/230905252/Lab10$ ./exe/q0
Enter input matrix A:
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
Enter input matrix B:
16 15 14 13
12 11 10 9
8 7 6 5
4 3 2 1
Resultant matrix:
80 70 60 50
240 214 188 162
400 358 316 274
560 502 444 386
Kernel execution time: 0.292608 ms
```

```
6CSEC2@debian:~/Documents/230905252/Lab10$ ./exe/q1
Enter number of rows for Mat A: 4
Enter number of cols for Mat A: 4
Enter number of rows for Mat B: 4
Enter number of cols for Mat B: 4
Enter input matrix A:
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
Enter input matrix B:
16 15 14 13
12 11 10 9
8 7 6 5
4 3 2 1
Resultant matrix:
80 70 60 50
240 214 188 162
400 358 316 274
560 502 444 386
Kernel execution time: 0.254976 ms
```

```
6CSEC2@debian:~/Documents/230905252/Lab10$ ./exe/q2
Enter size of vector: 7
Enter size of mask: 3
Enter vector N:      1 2 3 4 5 6 7
Enter mask M:        3 4 5
Convolutioned vector: 14 26 38 50 62 74 46
Kernel execution time: 0.072704 ms
```

```
6CSEC2@debian:~/Documents/230905252/Lab10$ ./exe/q3
Enter size of vector: 7
Enter size of mask: 3
Enter vector N:      1 2 3 4 5 6 7
Enter mask M:        3 4 5
Convolutioned vector: 14 26 38 50 62 74 46
Kernel execution time: 0.060416 ms
```

```
6CSEC2@debian:~/Documents/230905252/Lab10$ ./exe/q4
Enter number of friends: 3
Enter number of items: 3
Enter price of all items:
3 6 5
Enter quantity of items bought by friend 1:
1 0 2
Enter quantity of items bought by friend 2:
2 3 1
Enter quantity of items bought by friend 3:
1 1 3
Total purchase cost: 66
```